

Dipayan CHAKRABORTY

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Research domains: identification problems in graphs, dominating sets, graph modification problems, graph algorithms, computational & parameterized complexity

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Professional experience

- Aug. 2026 - till date **Assistant Professor, Department of Computer Science**, SRM University-AP, Amaravati, Andhra Pradesh, India
- Sep. 2025 - June 2026 **Assistant Professor (ATER in the French higher education system)**, Centrale Méditerranée, Marseille – Laboratoire d'Informatique et Systèmes (LIS), Aix-Marseille University, France
- Feb. 2025 - Aug. 2025 **Post-doctorate** (supervisor: **Faisal N. Abu-Khzam**), Department of Computer Science and Mathematics, Lebanese American University, Lebanon
- Jan. 2022 - Janv. 2025 **PhD scholar in Computer Science**, Laboratoire d'Informatique, de Modélisation et d'Optimisation des Systèmes (LIMOS), Université Clermont Auvergne, France
- Jan. 2023 - Jan. 2025 **Research associate** (academic host: **Michael Henning**), Department of Mathematics and Applied Mathematics, University of Johannesburg, South Africa
- Oct. 2022 - Nov. 2024 **Part-time lecturer**, Institut universitaire de technologie, Université Clermont Auvergne, France
- Jan. 2021 - Dec. 2021 **Contractual researcher**, as part of the project funded by SERB-SRG, Department of Computer Science and Engineering, Indian Institute of Technology Dharwad, Karnataka, India
- Aug. 2016 - Dec. 2020 **Assistant Professor**, Gargi Memorial Institute of Technology, Maulana Abul Kalam Azad University of Technology, Baraipur, Kolkata, India
- Jul. 2015 - Jul. 2016 **Visiting Lecturer**, Government College of Engineering and Ceramic Technology, Maulana Abul Kalam Azad University of Technology, Kolkata, India

Education and Trainings

- Jan. 2022 - Dec. 2024 **PhD in Computer Science** (defended 9th Dec. 2024)
Thesis title **Structural & algorithmic aspects of identification problems in graphs**, Laboratoire d'Informatique de Modélisation et d'Optimisation des Systèmes, Université Clermont Auvergne, France
- Sep. 2009 - Mar. 2012 **Master in mathematics** (ALGANT Erasmus Mundus)
Université de Bordeaux, France (1st year) – Università degli Studi di Padova, Italy (2nd year)
Master thesis **Mordell-Weil theorems & the Birch and Swinnerton-Dyer conjecture** (defended Mar. 2012), Università degli Studi di Padova, Italy
- Aug. 2004 - June 2007 **Bachelor of Science** (Honours in mathematics)
St. Xavier's College, Kolkata, University of Calcutta, Kolkata, India

(Selected) publications in journals (career total : 12)

- Published Sep. 2026 **Tight (double) exponential bounds for identification problems: locating-dominating set and test cover**, Dipayan Chakraborty, Florent Foucaud, Diptapriyo Majumdar and Prafullkumar Tale, *SIAM Journal on Discrete Mathematics* 40(3) (2026)

- Published Mar. 2026 **Identifying codes in triangle-free graphs of bounded maximum degree**, Dipayan Chakraborty, Florent Foucaud, Michael Henning and Tuomo Lehtilä, *Journal of Graph Theory* 112(3): 306-336 (2026)
- Published Mar. 2026 **Partitioning the vertex set of a graph into a dominating set and a locating dominating set**, Dipayan Chakraborty, Florent Foucaud, Michael Henning and Tero Laihonen, *The Electronic Journal of Combinatorics* 33(1): #P1.51 (2026)
- Published Jan. 2025 **A linear algorithm for radio k -coloring of powers of paths having small diameters**, Dipayan Chakraborty, Soumen Nandi, Sagnik Sen and Supraja DK, *Journal of Computer and System Sciences* 147: 103577 (2025)

(Selected) publications in conferences (career total : 12)

- Published June 2025 **Structural parameterization of locating-dominating set and test cover**, Dipayan Chakraborty, Florent Foucaud, Diptapriyo Majumdar and Prafullkumar Tale, *14th International Conference on Algorithms and Complexity (CIAC), Lecture Notes in Computer Science 15679: 187-204 (2025)*
- Published Dec. 2024 **Tight (double) exponential bounds for identification problems: Locating-Dominating Set and Test Cover**, Dipayan Chakraborty, Florent Foucaud, Diptapriyo Majumdar and Prafullkumar Tale, *35th International Symposium on Algorithms and Computation (ISAAC), Leibniz International Proceedings in Informatics 322: 19:1-19:18 (2024)*
- Published Sep. 2023 **Contracting edges to destroy a pattern: A complexity study**, Dipayan Chakraborty and Sandeep RB, *24th International Symposium on Fundamentals of Computation Theory (FCT), Lecture Notes in Computer Science 14292: 118-131 (2023)*
- Published June 2023 **A linear algorithm for radio k -coloring powers of paths having small diameter**, Dipayan Chakraborty, Soumen Nandi, Sagnik Sen and DK Supraja, *34th International Workshop on Combinatorial Algorithms (IWOCA), Lecture Notes in Computer Science 13889: 148-159 (2023)*

Research supervision

- May 2025 - Dec. 2025 **Intern of 3rd year bachelor's degree, PSG College of Technology, Coimbatore, India**,
Subject of research : *Identification problems in graphs*
Co-supervised with Prof. Florent Foucaud, Université Clermont Auvergne, France
- May 2024 - July 2024 **Intern of 3rd year bachelor's degree, Chennai Mathematical Institute, Chennai, India**,
Subject of research : *Local identification problems in graphs*
Co-supervised with Prof. Florent Foucaud, Université Clermont Auvergne, France

Organisation of scientific events

- Organising committee (Mar. 2024) **41st International Symposium on Theoretical Aspects of Computer Science, Clermont-Ferrand, France, (STACS 2024)**
Organised by: Université Clermont-Auvergne, France

Teaching activities

SRM University-AP, Amaravati, Andhra Pradesh, India

- Aug. 2026 - till date **Computer Science and Engineering (Track AI & ML): B. Tech 2 (2026-2027 || 225 hours)**
• Design and analysis of algorithms

Centrale Médiétrannée, Château-Gombert, Marseille, France

- Sep. 2025 - June 2026 **Computer Science Engineering : Bachelor year 1 (176 hours)**
• JAVA • Algorithms • Database management

- Nov. 2025 - Jan. 2026 **Computer Science Engineering : Bachelor year 2 (18 hours)**
• Scientific Python • Numpy • Matplotlib • Pandas • Seaborn • Scikit

Aix-Marseille Université, Saint-Jerôme, Marseille, France

- Jan. 2025 - till date **Bachelor of Science : year 1 (30 hours)**
• Programming • Algorithms

Oct. 2025 - Dec. 2025 **Bachelor of Science : year 2** (10 hours)

• Graph theory • Trees • Algorithms

IUT Clermont Auvergne, *Université Clermont Auvergne*, Aubière, France

2022 - 2024 **Bachelor Universitaire de Technologie (BUT) : year 2** (42 hours)

• Methods of optimisation

Gargi Memorial Institute of Technology (GMIT), *MAKAUT*, Kolkata, India

2016 - 2020 **Bachelor of Technology : year 1 to year 4** (861 hours)

• Graph theory & algorithms • Linear algebra • Abstract algebra • Probability • Real analysis • Complex analysis • Numerical methods • Differential calculus • Integral transforms

Government College of Engineering & Ceramic Technology, *MAKAUT*, Kolkata, India

2015 - 2016 **Bachelor of Technology : year 1 & year 2** (77 hours)

• Probability • Integral transforms • Fourier analysis • Differential equations